Republic of the Philippines
BATANGAS STATE UNIVERSITY
The National Engineering University
Alangilan Campus
Golden Country Homes, Alangilan Batangas City, Batangas, Philippines 4200
Tel Nos.: (+63 43) 425-0139 local 2222 / 2221
E-mail Address: cics.alangilan@g.batstate-u.edu.ph | Website Address: http://www.batstate-u.edu.ph

College of Informatics and Computing Sciences

PROJECT PROPOSAL
OOP 101 – OBJECT-ORIENTED PROGRAMMING
Computer Science Department
1st Semester, AY 2026-2027

I. TEAM INFORMATION

Team Name:	Yearn Inc.		
Team Number:	11	**Section:**	CS-2102

TEAM MEMBERS			
No.	**SR-Code**	**Name**	**GitHub Username**
1	25-02031	Garcia, Richmond Flavian E.	garciarichmondflavian-eng
2	25-00451	Magtibay, Khryss Dilwyn S.	Khryss19
3	25-03285	Saguid, Kristine P.	eskups04

II. PROJECT INFORMATION

Project Title
Pa-Ihee

Project Background

The school does not have a way of informing people which comfort rooms (CRs) are working and which are under maintenance. As a result, students and employees may go to a CR only to find it closed for repair or cleaning. This wastes time and causes inconvenience, especially during short breaks.

Maintenance staff also face a problem. Without a proper record, it is difficult to know which CRs need repair, what the problem is, and when each one was last checked.

This project aims to address these problems through a system that records and shows the status of each CR. It is important because it helps people save time and allows maintenance staff to manage repairs in an organized manner.

Project Objectives
General Objective
To develop a Java-based application that will help users check the availability of the stalls of the comfort room

Specific Objectives
The project aims to:
- Keep a list of comfort rooms and their locations
- Show whether the comfort room is working or under maintenance

- Allow staff to update the status of the comfort room and add a short reason
- Record the date and time for each update
- Keep a history of maintenance for each comfort room

Target Users

Target User	How They Will Use the System
Students	View the list of comfort rooms and their locations.
Maintenance Staff	Update the status of the comfort room, add a reason, and record repairs.
School Administrator	Review the maintenance history to monitor the condition of the comfort rooms.

Proposed Features

Feature	Description
Search and Filter	Find comfort rooms by floor, building, or status (for example, show only working CRs).
Nearest working CR	Suggest the closest working comfort room based on the user's floor or building.
Report a problem	Students and employees can report issues such as a broken faucet, no water, or a bad smell.
Availability	Mark a comfort room whether it is working or under maintenance
Login with roles	Students, staff, and administrators see different menus.

III. JAVA AND OOP CONCEPTS

OOP Concept	How It Will Be Used
Classes and Objects	Each comfort room, stall, and urinal is stored as its own item with its own status.
Methods	Actions such as viewing the list, changing a status, and counting working stalls each have their own set of instructions.
Encapsulation	The status of a stall or urinal is protected so it can only be changed through the proper update process.
Inheritance	Stalls and urinals share the same basic details, such as an ID and status. Each one builds on these with

	its own details. Students and staff also share basic user details.
Polymorphism	Stalls and urinals can show their details in their own way, while the system treats them in the same way. The menu also changes depending on the type of user.
Abstraction	The system defines what every fixture or user must have, without describing a specific one.
Exception Handling	If a user enters an invalid choice, the system shows a message and asks again instead of stopping.
Arrays/Data Handling	Lists are used to store comfort rooms, their stalls and urinals, and maintenance records.

IV. PROPOSED CLASSES

Class	Purpose	Main Attributes	Main Methods
Main System	Runs the program and shows the main menu.	List of comfort rooms	Start the system, show the menu, exit
ComfortRoom	Represents one comfort room.	Name, type (Women, Men, or All Gender), list of fixtures	Count working fixtures, get overall status, show detail
Fixture	The basic type for anything inside a comfort room that can be used or repaired.	ID, status, maintenance note, last updated	Change status, add note
Stall	A toilet stall. Based on Fixture.	Stall number	Show details
Urinal	A urinal. Based on Fixture.	Urinal number	Show details
User	The basic type of user.	Name, role	Show menu

Student/Faculty/ Maintenance Staff	Specific users. Students and employees can only view. Staff can also update statuses.	Based on User	View status, update status, view history
MaintenanceRecord	Stores one past update or repair.	Comfort room, fixture, reason, data and time	Show record

The classes may be changed as the project is developed.

V. PROGRAM FLOW

The user starts the system and selects their role from the main menu. A student or employee can view the comfort rooms and see how many stalls and urinals are working. Maintenance staff can choose a comfort room, select a stall or urinal, change its status, and enter a reason. The system checks the input, saves the update with the date and time, and shows the new status. The user then returns to the main menu.

Start
↓
Main Menu
↓
Select User Role
↓
Select Operation (View/ Update/ View History)
↓
Enter Information (Floor, Comfort Room, Stall or Urinal, Status, Reason)
↓
Validate Input
↓
Process Request
↓
Display Result
↓
Return to Main Menu
↓
Exit

VI. TEAM RESPONSIBILITIES

Member	Assigned Tasks

Garcia, Richmond Flavian E.	Lead Programmer
Magtibay, Khryss Dilwyn S.	Project Leader/ System/Design Lead
Saguid, Kristine P.	Documentation/Support Programmer

VII. DEVELOPMENT SCHEDULE

Week	Activity	Expected Output
Week 1	Planning and requirements	Project proposal
Week 2	System and class design	Class/UML diagram
Week 3	Initial coding	Basic classes
Week 4–5	Feature development	Working features
Week 6	OOP implementation	Completed OOP features
Week 7	Testing and debugging	Tested application
Week 8	Documentation and presentation	Final project

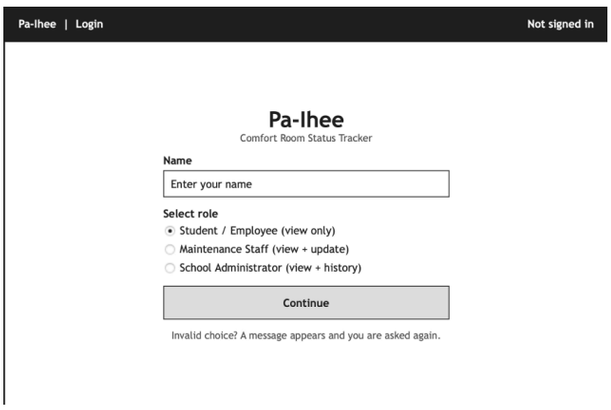
The schedule may be adjusted based on the course calendar.

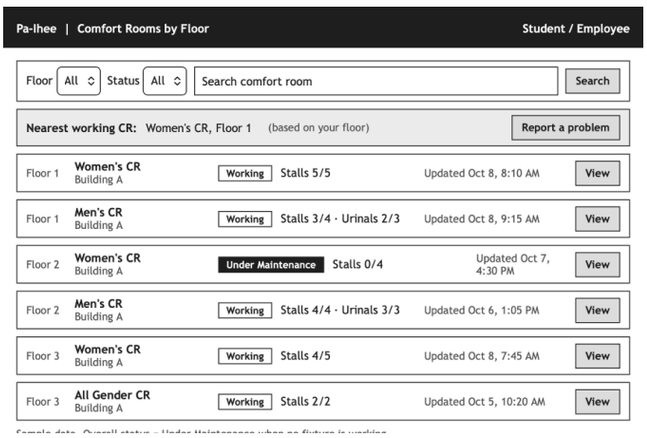
VIII. EXPECTED OUTPUT

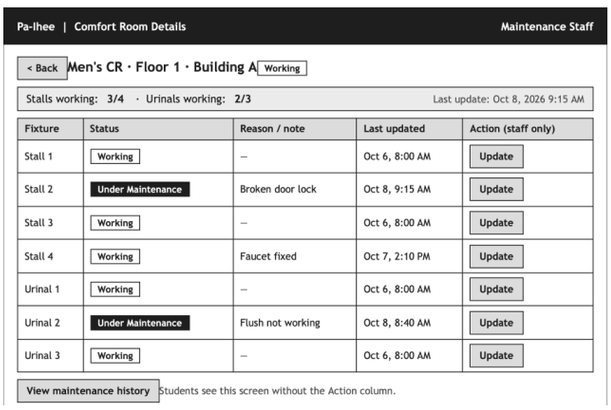
The team expects to produce a functional Java application that shows whether each comfort room, stall, and urinal is working or under maintenance and keeps a record of updates. The application will demonstrate the appropriate use of Java and object-oriented programming concepts.

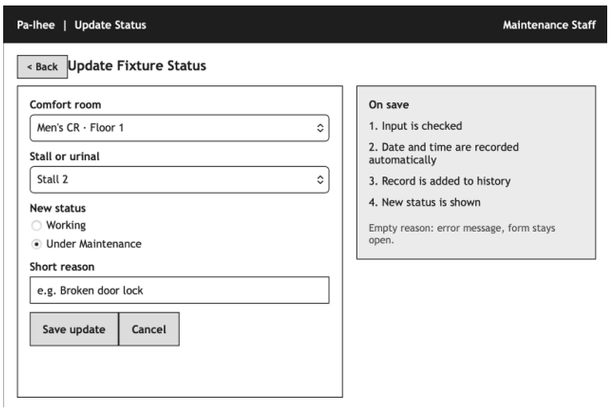
When finished, the system should let users:

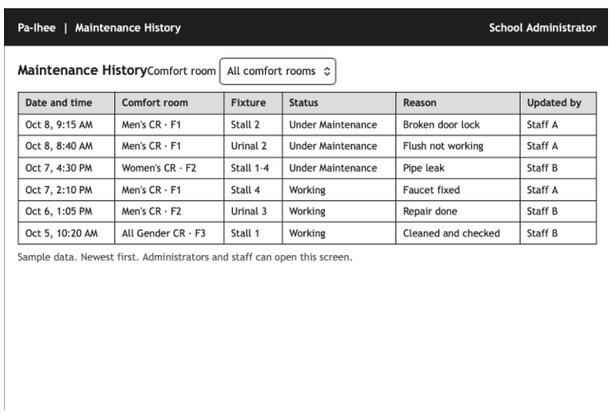
- View all six comfort rooms by floor and see their current status
- See how many stalls and urinals in each comfort room are working.
- Update the status of the stall or urinal with a short reason (staff only)
- See the date and time of the latest update
- Review the maintenance history of each comfort room











IX. PROJECT LIMITATIONS

- The system does not send notifications, alarms or messages to the users or staff.
- The system does not show whether the stall is vacant or occupied.
- The system does not track water supply, cleanliness and supplies such as soap and tissue.

X. GITHUB REPOSITORY

Repository Name:
Link to Repository:

XI. APPROVAL

Prepared by:

Name	Signature
Garcia, Richmond Flavian E	
Magtibay, Khryss Dilwyn S.	
Saguid, Kristine P.	

Instructor's Comments:

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Project Status:

☐ Approved
☐ Approved with Revisions
☐ Revise and Resubmit

Name of Instructor:

Ms. FATIMA MARIE P. AGDON, MSCS

Date Signed: _____

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